AMINU MUSA AMBURSA

Graduate Data Scientist | Data Analyst

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PROFILE

MSc Data Science student at the University of Exeter with a BSc in Computer Science. Project experience in machine learning, NLP, SQL and computer vision, with reproducible Python workflows. Seeking a graduate or junior data role.

SKILLS

- **Programming and data:** Python, SQL (MySQL), data preprocessing, EDA, visualisation
- **Machine learning:** Linear, Random Forest and Support Vector Regression; K-Means; MLP neural networks (PyTorch); hyperparameter tuning
- **Evaluation:** R^2, MAE, RMSE, ROC-AUC, mAP, precision/recall, bootstrap confidence intervals
- **NLP and networks:** sentiment classification, topic modelling, graph features
- **Computer vision:** object detection (YOLOv8)
- **Workflow:** modular code, testing, DVC, HTML reporting · **Also:** cloud and network security fundamentals

EDUCATION

MSc Data Science, University of Exeter, UK — 2026 – present
BSc Computer Science, University of Science and Technology Aliero, Nigeria — 2016 – 2021

SELECTED PROJECTS

California Housing Price Prediction · Machine Learning

- Built a leakage-free pipeline on 20,640 districts: model-based imputation (RMSE 80 vs 410 for median fill), ratio features and K-Means (k = 6) geographic clusters.
- Tuned and compared Linear, Random Forest, SVR and MLP models. **Random Forest was best (R^2 0.851, test RMSE ≈ $44.0k)**; the MLP followed (R^2 0.818).
- Showed with bootstrap tests that the model rankings were significant ($p < 0.001$).

Neural Network Regression (MLP)

- Selected the architecture by random search over 30 configurations (4 hidden layers, dropout 0.2, early stopping).
- Regularisation cut the train–test RMSE gap from ≈$21.0k to ≈$2.0k, giving the best generalisation of the four models.

Amazon Digital Music Review Sentiment · NLP and Network Analysis

- Classified sentiment on 123,994 reviews using TF-IDF, VADER, LDA topics and product co-review network features.
- Text-only ROC-AUC was 0.946. Adding topic and network features raised it to 0.951, and the network features added a statistically significant signal (McNemar $p < 0.001$).

Object Detection with YOLOv8 · Computer Vision

- Detected cars and pedestrians with YOLOv8n on 100 images (45 COCO, 45 BDD100K, 10 self-captured).
- Results: mAP@0.5 0.523, precision 0.687, recall 0.512, 0.15 s per image on CPU. Error analysis showed failures on small, occluded, low-light objects, with image enhancement proposed as a fix.

Farm Sustainability Database · SQL

- Normalised a 23-column flat file to 3NF across six tables, with an ERD, and implemented it in MySQL with keys, constraints and indexes.
- Verified the schema with joins that reproduced all 20 source rows. Also designed a RESTful API and a MongoDB document alternative.

Reproducible Data Science Pipeline

- Python workflow covering classification, clustering and regression, with modular functions, tests, DVC stages, configurable parameters and HTML reports.

CERTIFICATIONS

- Certificate of Cloud Security Knowledge (CCSK)
- Cisco Networking Academy – Introduction to Cybersecurity
- Trend Micro CyberTalents – Introduction to Cybersecurity Bootcamp
- Mastercard Foundation – Jobberman Accelerated Soft Skills Training

MEMBERSHIPS

Exeter Computer Science Society · IEEE Computer Society · British Computer Society · Nigeria Computer Society (Student Member)

LANGUAGES

English – full professional proficiency